



University
of Exeter

Social Networks & Text Analysis (ECMM447J)

Dr. Rodrigo Wilkens

Intended Learning Outcomes

- Understand the strengths, weaknesses and biases of different NLP methods
- Clean, parse, transform and process large collections of text documents
- Use modern NLP techniques to derive insight from unstructured data

Outline

- Recap & Motivation
- Lexical Semantics
- NER
- Sentiment and Emotions
- Topic Modeling

Outline

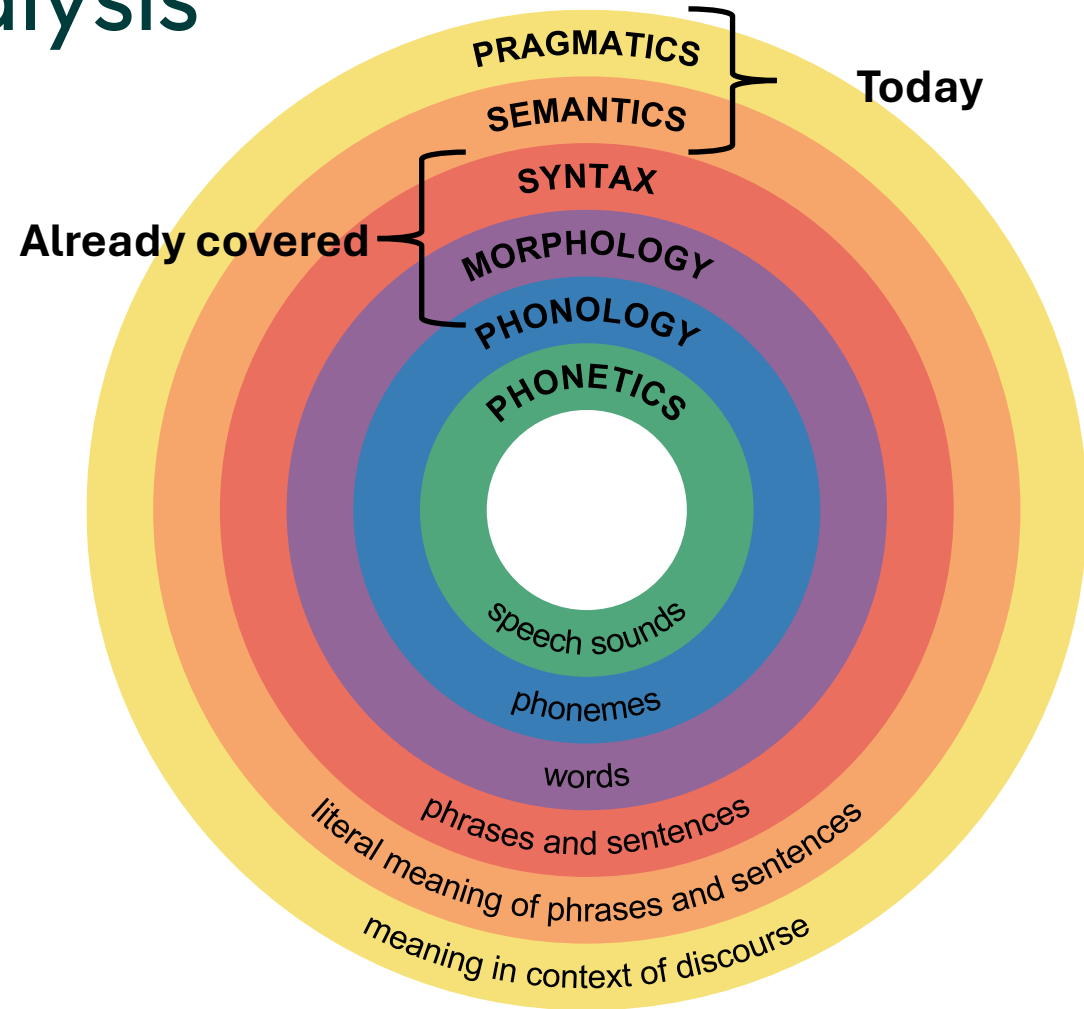
- **Recap & Motivation**
- Lexical Semantics
- NER
- Sentiment and Emotions
- Topic Modelling

Recap & Motivation

- Last week: ILO
 - Understand the strengths, weaknesses and biases of different NLP methods
 - Compare frequency-, statistical-, and sequence-based methods in terms of expressiveness, limitations, and bias.
 - Clean, parse, transform and process large collections of text documents
 - Apply weighting, keyword extraction, and syntactic annotation to structured text representations.
 - Know about recent advances and applications of NLP methods, their impact and real-world applications
 - Explain methods of keyword extraction, PoS tagging, and parsing
 - Use modern NLP techniques to derive insight from unstructured data
 - Demonstrate methods of language understanding.

From strings to data structures

- Types of linguistic analysis



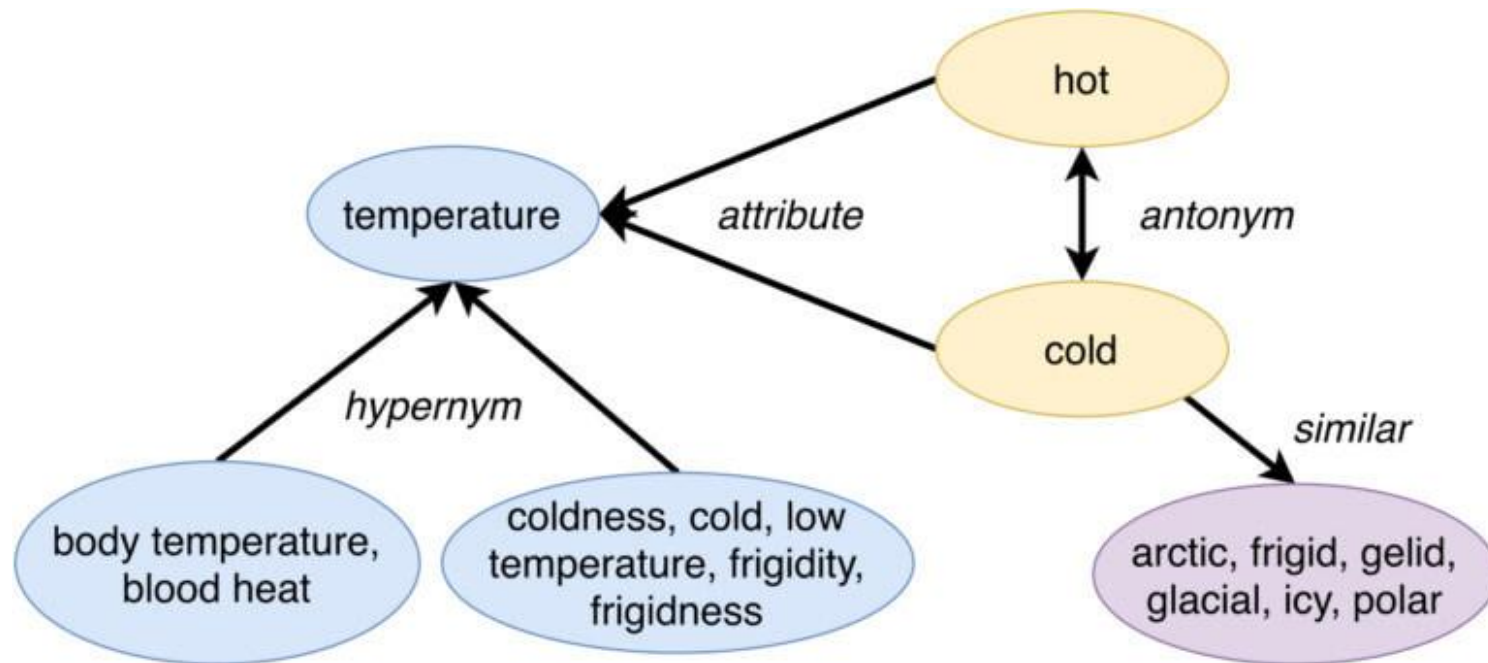
(Pragmatics or Discourse)

Outline

- Recap & Motivation
- **Lexical Semantics (short intro)**
- NER
- Sentiment and Emotions
- Topic Modelling

Lexical Semantics

- Lexical databases: Wordnet

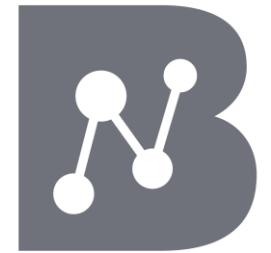
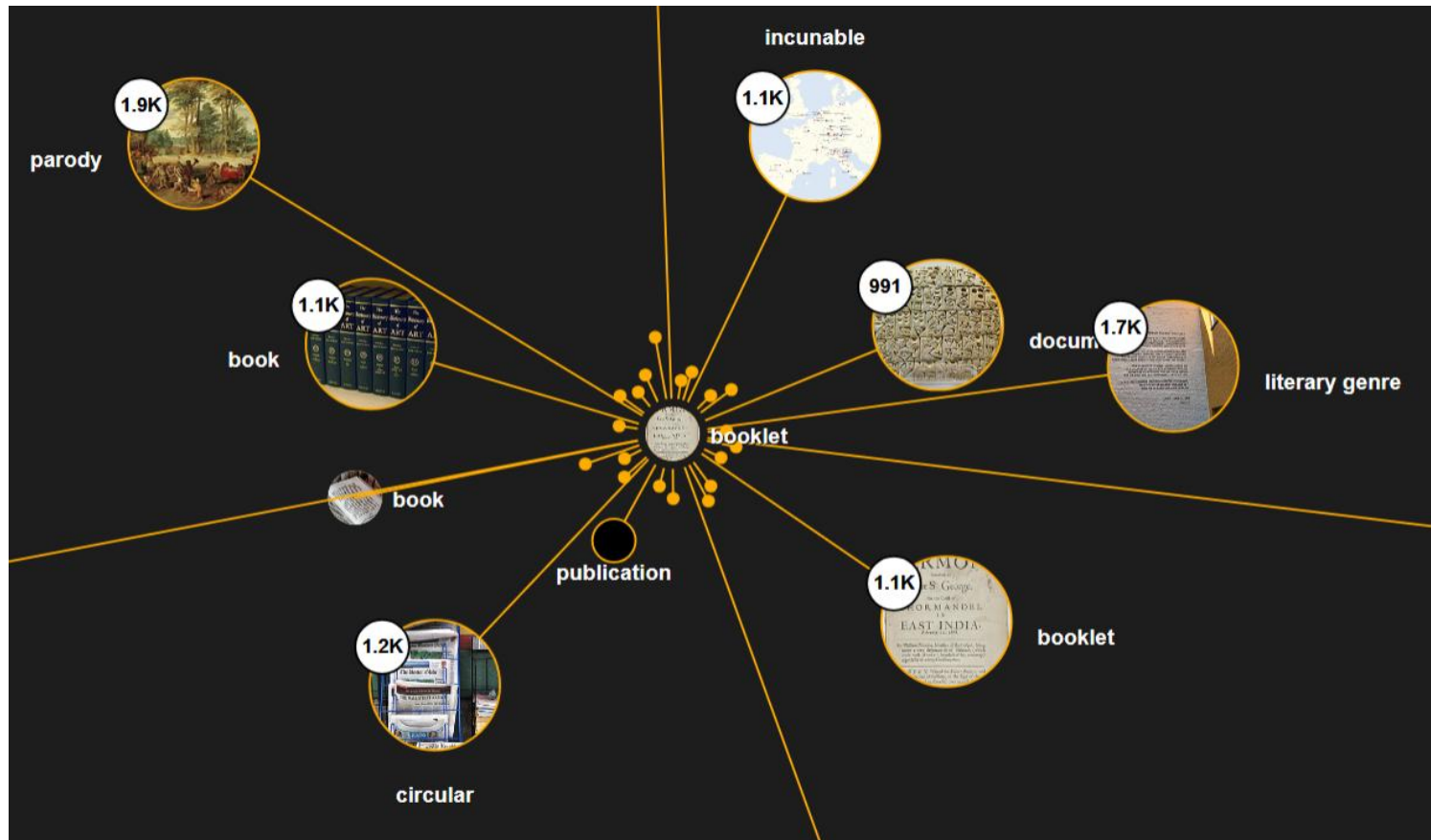


Lexical Semantics

- What are the synonyms, antonyms and hypernyms of *cold*?
 - [menti.com](https://www.menti.com)

Lexical Semantics

- Lexical databases: BabelNet



BabelNet

<https://babelnet.org/>

Outline

- Recap & Motivation
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Named Entity Recognition (NER)

- A named entity is, roughly speaking, anything that can be referred to with a proper name: a person, a location, an organisation.

Type	Tag	Sample Categories	Example sentences
People	PER	people, characters	Turing is a giant of computer science.
Organization	ORG	companies, sports teams	The IPCC warned about the cyclone.
Location	LOC	regions, mountains, seas	Mt. Sanitas is in Sunshine Canyon .
Geo-Political Entity	GPE	countries, states	Palo Alto is raising the fees for parking.

Figure 17.5 A list of generic named entity types with the kinds of entities they refer to.

Example of Extracted Structure

- Input sentence:
 - Theresa May visited Brussels on Monday.
- Extracted entities:
 - PERSON: *Theresa May*
 - LOCATION: *Brussels*
 - DATE: *Monday*
- Extracted relation:
 - *visited(PERSON, LOCATION)*

Examples of Named Entities?

- Common categories
 - person, organisation, location, date, time, money, percent, product and event
- Biomedical categories
 - gene, disease and drug

Examples of Named Entities?

- Challenges:
 - single-word and multiword entities
 - ambiguous entities
- Linguistic Complexity in Entities
 - US / U.S. / United States
 - COVID-19 / Covid 19 / SARS-CoV-2 disease
- Nested entities:
 - [[University of Exeter] [Computer Science]]

Gazetteers (Entity Dictionaries)

- Examples:
 - City lists, Airports, Diseases, Companies, Political parties, Drug names
- Advantages:
 - high precision
 - simple to update in small domains
- Problems:
 - Outdated entries
 - Ambiguity
 - no handling of unseen entities

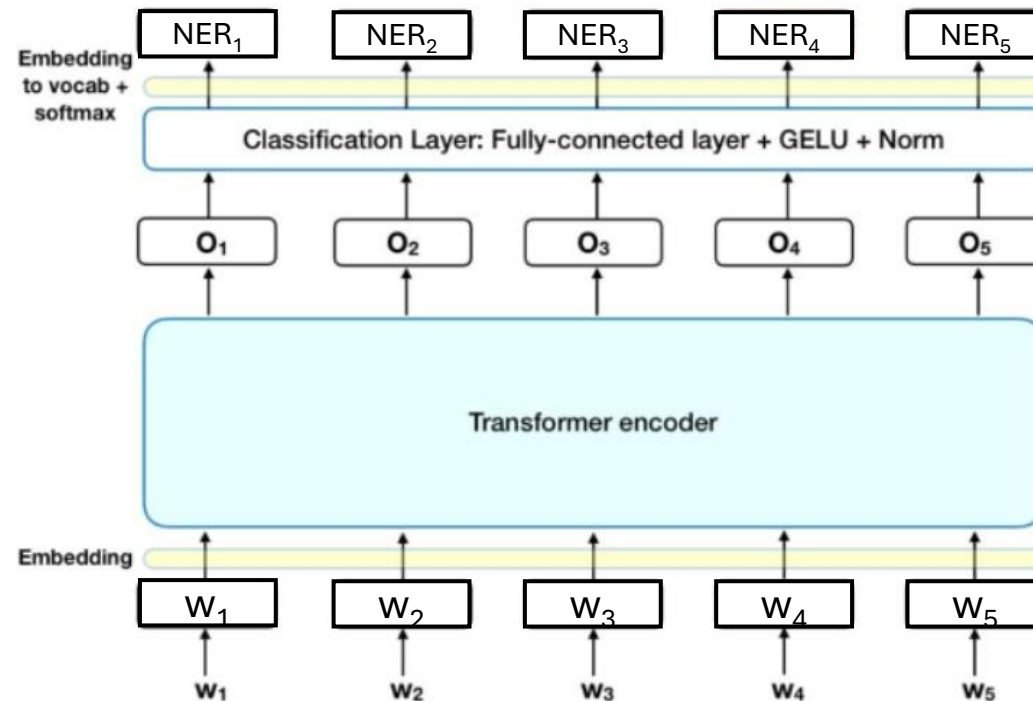
Gazetteers (Entity Dictionaries)



- Let's create a list of cities
 - [menti.com](https://www.menti.com)

Named Entity Recognition (NER)

- How to NER:
 - Specialised models



Named Entity Recognition (NER)

- Challenge: multiple tokens

Words
Jane
Villanueva
of
United
Airlines
Holding
discussed
the
Chicago
route
.

Named Entity Recognition (NER)

- BIO Tagging Scheme

- Tag definitions:

- B-XXX: beginning of entity
 - I-XXX: inside entity
 - O: outside entity

- Purpose:

- enforce valid entity spans

- Issues

- Nested entities: [[University of Exeter] [Computer Science]]
 - Exeter can be I and O

Named Entity Recognition (NER)

- BIOES: More Precise Boundaries
 - Begin – Inside – Outside – End – Single
- Common errors:
 - over- and under-extended spans
 - nested conflicts
- Example:
 - University/**B** of/**I** Exeter/**E** Computer/**I** Science/**E**

Named Entity Recognition (NER)

- Challenge: multiple tokens

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Figure 17.7 NER as a sequence model, showing IO, BIO, and BIOES taggings.

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- Recap & Motivation
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- NER
- **Sentiment and Emotions**
- Topic Modelling

Sentiment and Emotions

- Sentiment

Positive	Negative
GREAT movie and the family will love it!! If kids are bored one day just pop the tape in and you'll be so glad you did!!! ~Rube i luv raven-s!	The script for this movie was probably found in a hair-ball recently coughed up by a really old dog. Mostly an amateur film with lame FX. For you Zeta-Jones fanatics: she has the credibility of one Mr. Binks.
Did Sandra (yes, she must have) know we would still be here for her some nine years later? See it if you haven't, again if you have; see her live while you can.	I would love to have that two hours of my life back. It seemed to be several clips from Steve's Animal Planet series that was spliced into a loosely constructed script. Don't Go, If you must see it, wait for the video ...
Verry classic plot but a verry fun horror movie for home movie party Really gore in the second part This movie proves that you can make something fun with a small budget. I hope that the director will make another one	This is without a doubt the worst movie I have ever seen. It is not funny. It is not interesting and should not have been made.

Sentiment and Emotions

- Lexical methods for sentiment analysis rely on lists of words with associated polarity scores
(manually labelled by many people typically!)
- Polarity usually is either positive or negative (happy vs sad, ...)
 - “I hate swimming”
 - “This film is amazing!”
- Modern approaches rely on machine learning:
 - neural networks

Sentiment and Emotions

- **VADER** *Valence Aware Dictionary and sEntiment Reasoner* is a lexicon and rule-based sentiment analysis tool
 - Specifically designed for social media!
 - 10 independent human raters manually rated over 9,000 token features for sentiment polarity and intensity (combined in a compound score -1 to 1)
 - Includes Western-style emoticons and acronyms

Sentiment and Emotions

- **textblob** *Simplified Text Processing*
 - Another lexicon-based sentiment analysis method
 - Includes polarity and subjectivity

Sentiment and Emotions

- Emotions
 - Many emotions can be expressed in text (*but not only*)
 - Universal emotions?



Sentiment and Emotions

- Emotions
 - LIWC** (Linguistic Inquiry and Word Count) has over 100 built-in dictionaries to capture people's social and psychological states

	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU
1																				
2	121		122		123		124		125				126			127			128	
3	Social		Family		Friends		Humans		Affect				Posemo			Negemo			Anx	
4	acquainta	gives	persons	aunt	acquainta	adult	abandon	damn	fume	kindn	privileg	supporting	accept	freed	partie	abandon	enrag	maddening	snob	afraid
5	admit	giving	person's	bro	amigo	adults	abuse	danger	fuming	kiss	prize	supportive	accepta	freeing	party	abuse	envie	madder	sob	alarm
6	admits	gossip	persua	bros	bf	babe	abusi	daring	fun	laidback	problem	supports	accepted	freely	passion	abusi	envious	maddest	sobbed	anguish
7	admitted	grandchil	phone	brother	boyf	babies	accept	darlin	funn	lame	profit	suprem	accepting	freeness	peace	ache	envy	maniac	sobbing	anxi
8	admitting	granddad	phoning	cousin	bud	baby	accepta	daze	furious	laugh	promis	sure	accepts	freer	perfect	aching	evil	masochis	sobs	apprehens
9	adult	granddau	prais	dad	buddies	bambino	accepted	dear	fury	lazie	protest	surpris	active	frees	play	advers	excruciat	melanchol	solemn	asham
10	adults	grandf	private	daughter	buddy	boy	accepting	decay	geek	lazy	protested	suspicio	admir	friend	played	afraid	exhaust	mess	sorrow	aversi
11	advice	grandkid	provide	ex	colleague	boy's	accepts	defeat	genero	liabilit	protesting	sweet	ador	fun	playful	aggravat	fail	messy	sorry	avoid
12	advis	grandm	public	exes	companion	boys	ache	defect	gentle	liar	proud	sweetheart	advantag	funn	playing	aggress	fake	miser	spite	awkward
13	affair	grandpa	question	exhubby	companions	chick	aching	defenc	gentler	libert	puk	sweetie	adventur	genero	plays	agitat	fatal	miss	stammer	confus
14	amigo	grandson	reassur	exhusband	comrad	chick	active	defens	gentlest	lied	punish	sweetly	affection	gentle	pleasant	agoniz	fatigu	missed	stank	craz
15	anybod	granny	receiv	exwife	exbf	chicks	admir	definite	gently	lies	radian	sweetness	agree	gentler	please	agony	fault	misses	startl	desperat
16	anyone	group	refus	exwife	exboyfriend	child	ador	definitely	gigg	like	rage	sweets	agreeab	gentlest	pleasing	alarm	fear	missing	steal	discomfort
17	apolog	grownup	relationship	families	exgf	children	advantag	degrad	giver	likeab	raging	talent	agreed	gently	pleasur	alone	feared	mistak	stench	distract
18	argu	grudge	relatives	family	exgirl	child's	adventur	delectabl	giving	liked	rancid	tantrum	agreeing	gigg	popular	anger	fearful	mock	stink	distraught
19	armies	guest	reple	father	fellow	citizen	advers	delicate	glad	likes	rape	tears	agreement	giver	positiv	angr	fearing	mocked	strain	distress
20	army	guy	reply	folks	fiance	citizen	affection	delicious	gladly	liking	raping	teas	agrees	giving	prais	anguish	fears	mockers	strange	disturb
21	ask	he	request	grandchil	friend	citizens	afraid	deligh	glamor	livel	rapist	tehe	alright	glad	precious	annoy	feroc	mocking	stress	doubt
22	asked	hear	respond	granddad	gf	female	aggravat	depress	glamour	LMAO	readiness	temper	amaz	gladly	prettie	antagoni	feud	mocks	struggl	dread
23	asking	heard	role	granddau	girlfriend	gentlem	aggress	depriv	gloom	LOL	ready	temper	amor	glamor	pretty	anxi	fiery	molest	stubborn	dwel

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Topic Modelling



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Meaning of **topic** in English



topic

noun [C]

UK /'tɒp.ɪk/ US /'tɑː.pɪk/

Add to word list

B1

a subject that is discussed, written about, or studied:

- *Our discussion included topics such as acid rain and the melting of polar ice.*
- *Your essay can be on any topic you like.*

(Cambridge English Dictionary)

Topic Modelling

- How can we identify the topic of a text?

The heat generated by a washing-machine-sized data centre is being used to heat a Devon public swimming pool.

The computers inside the white box are surrounded by oil to capture the heat - enough to heat the pool to about 30C 60% of the time, saving Exmouth Leisure Centre thousands of pounds.

The data centre is provided to the council-run centre free of charge. Start-up Deep Green charges clients to use its computing power for artificial intelligence and machine learning.

Founder Mark Bjornsgaard said the company would also refund the leisure centre's electricity costs for running the "digital boiler" - and seven other England pools had signed up to the scheme.

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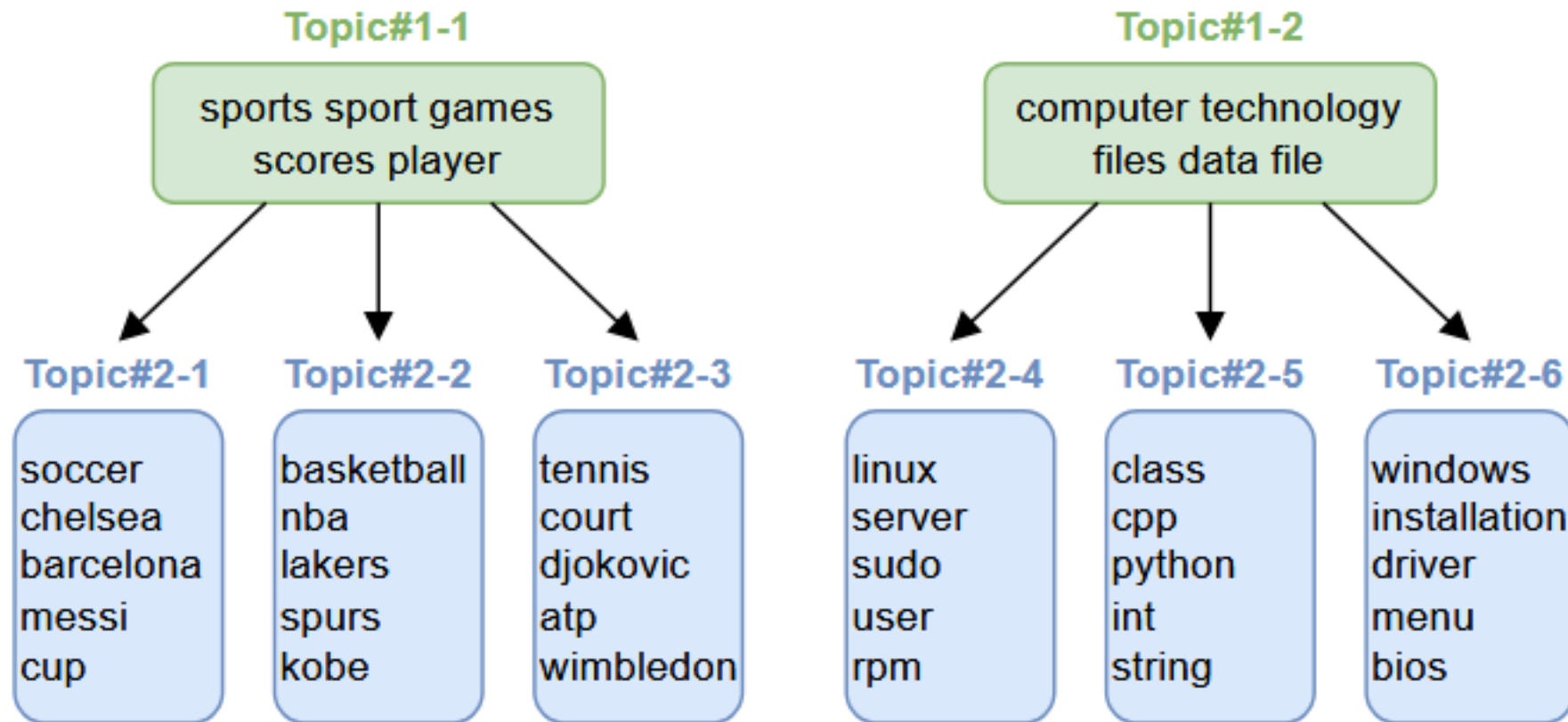
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Topic Modelling

- Topic as a set of words
 - **cat**: {miaow, fur, claws, stroke, feline, pet, animal}
 - **trucks**: {engine, road, wheels, cargo, driver, lorry}
 - **politics**: {party, issue, election, vote, economy, candidate}

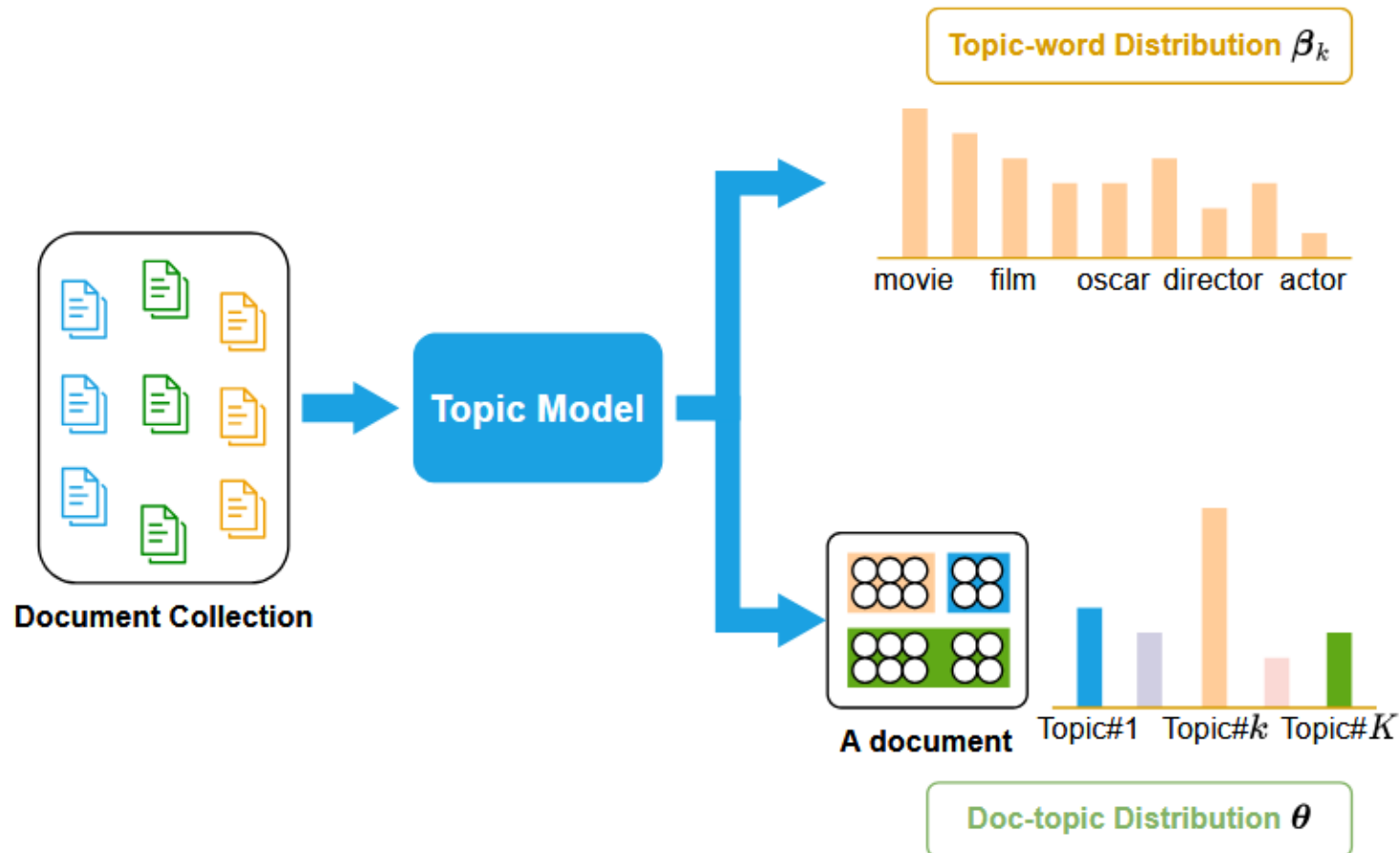
Topic Modelling

- Topic as a set of words



Topic Modelling

- Topic as a set of words



Topic Modelling

- Different approaches:
 - Latent Dirichlet Allocation (LDA)
 - Guided
 - Supervised
 - LLM

Topic Modelling

- Latent Dirichlet Allocation (LDA)
 - Assume a collection of **D** documents: $\{d_1, d_2, d_D\}$

Dog breeds vary widely in shape, size, and color. They perform many roles for humans, such as hunting, herding, pulling loads, protection, assisting police and the military, companionship, therapy, and aiding disabled people. This influence on human society has given them the sobriquet of "man's best friend".

d1

Cats are commonly kept as house pets but can also be farm cats or feral cats; the feral cat ranges freely and avoids human contact. Domestic cats are valued by humans for companionship and their ability to kill rodents. About 60 cat breeds are recognized by various cat registries.

d2

We're a nation of pet lovers, and rightly so! Cats and dogs offer unconditional love, affection and friendship, as well as a host of therapeutic benefits, such as improved general health and wellbeing - after all, who doesn't like a cuddle from a furry friend at the end of a long day?

d3

- Assume a collection of **K** topics exists: $\{k_1, k_2, \dots, k_K\}$



DOGS k_1



CATS k_2



PEOPLE k_3



CARS k_4

Topic Modelling: LDA



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d3



DOGS k1



CATS k2

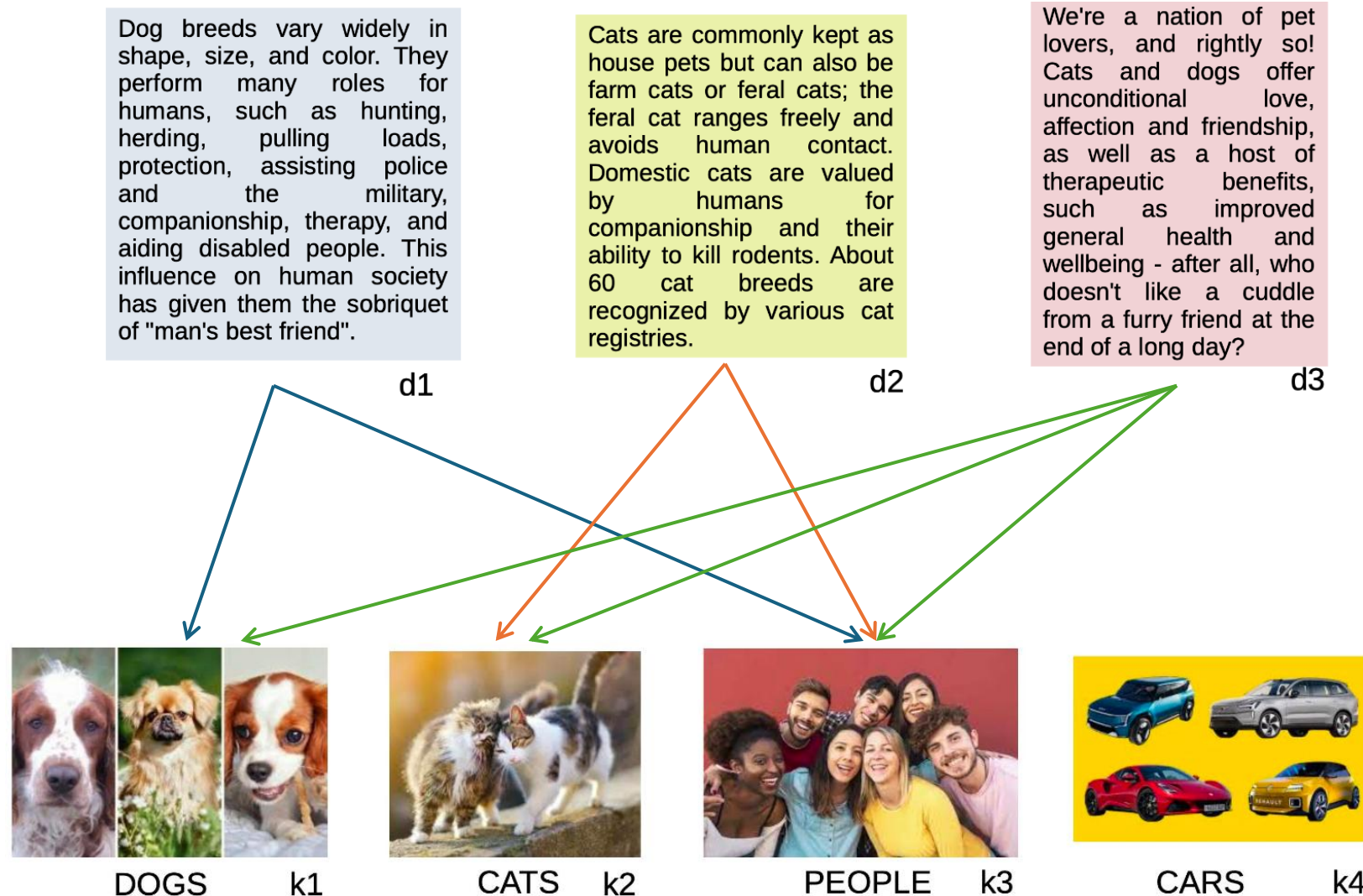


PEOPLE k3

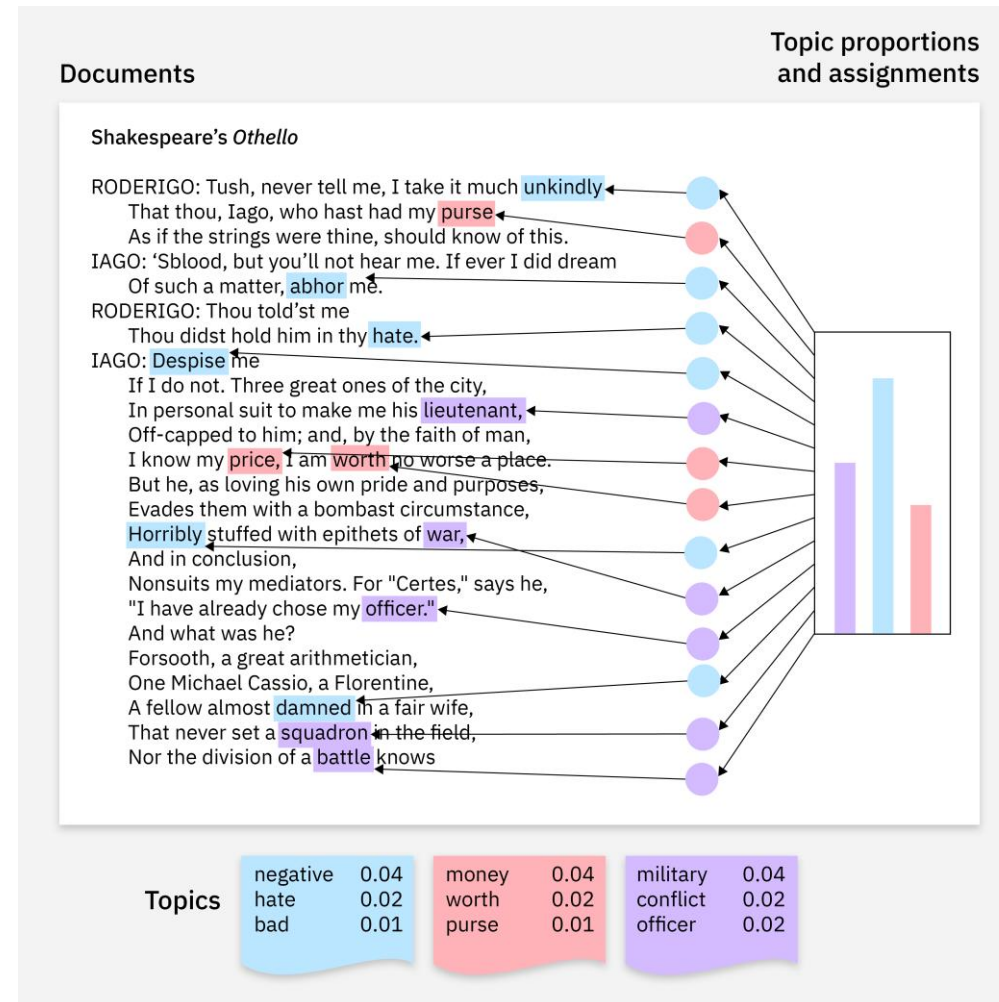


CARS k4

Topic Modelling: LDA

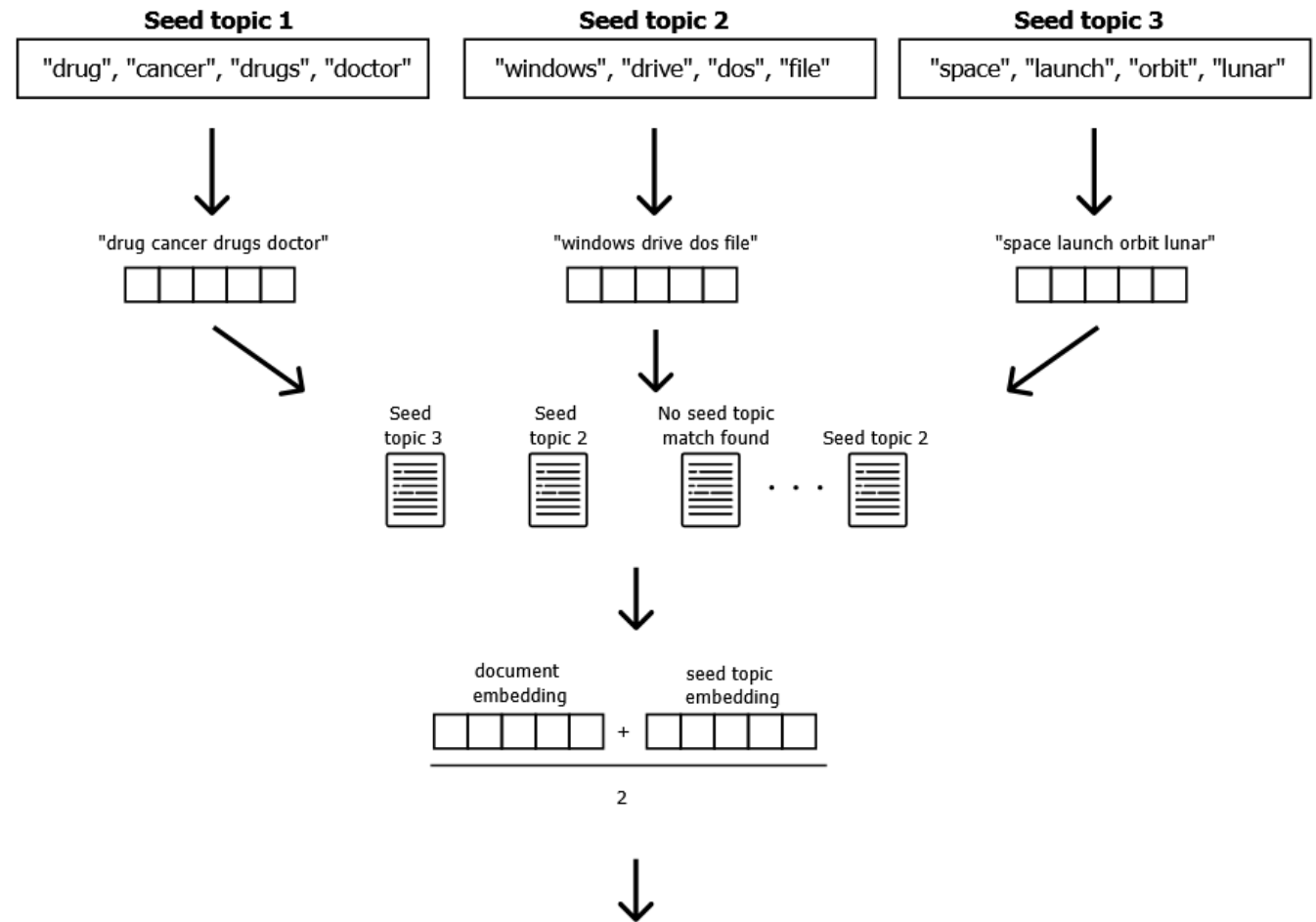


Topic Modelling: LDA



Topic Modelling

- Guided



Conclusion

- By the end of this session, students will be able to:
 1. Explain how lexical semantic resources (e.g. synonymy, hypernymy, semantic networks) support meaning representation in NLP, and identify their strengths and limitations.
 2. Describe and compare different approaches to Named Entity Recognition (gazetteer-based, sequence labelling), including their biases, failure cases, and domain limitations.
 3. Apply sequence labelling schemes (BIO, BIOES) to encode entity boundaries and reason about common boundary and ambiguity errors in NER outputs.
 4. Analyse the challenges posed by ambiguity, multi-word expressions, and nested entities, and assess how modelling choices affect extracted structures.
 5. Distinguish between lexicon-based and machine-learning-based approaches to sentiment and emotion analysis, including their assumptions, coverage, and sources of bias.
 6. Use NLP tools for entity recognition, topic modelling, sentiment/emotion analysis to derive structured information from text, and interpret their outputs critically.